

Iberian Cantilevers

Overhead line equipment from the Iberian Peninsula
for *Create: Pantographs & Wires*

Feature overview

Minecraft 1.20.1 (Forge) · version 0.1.0 · October 2026

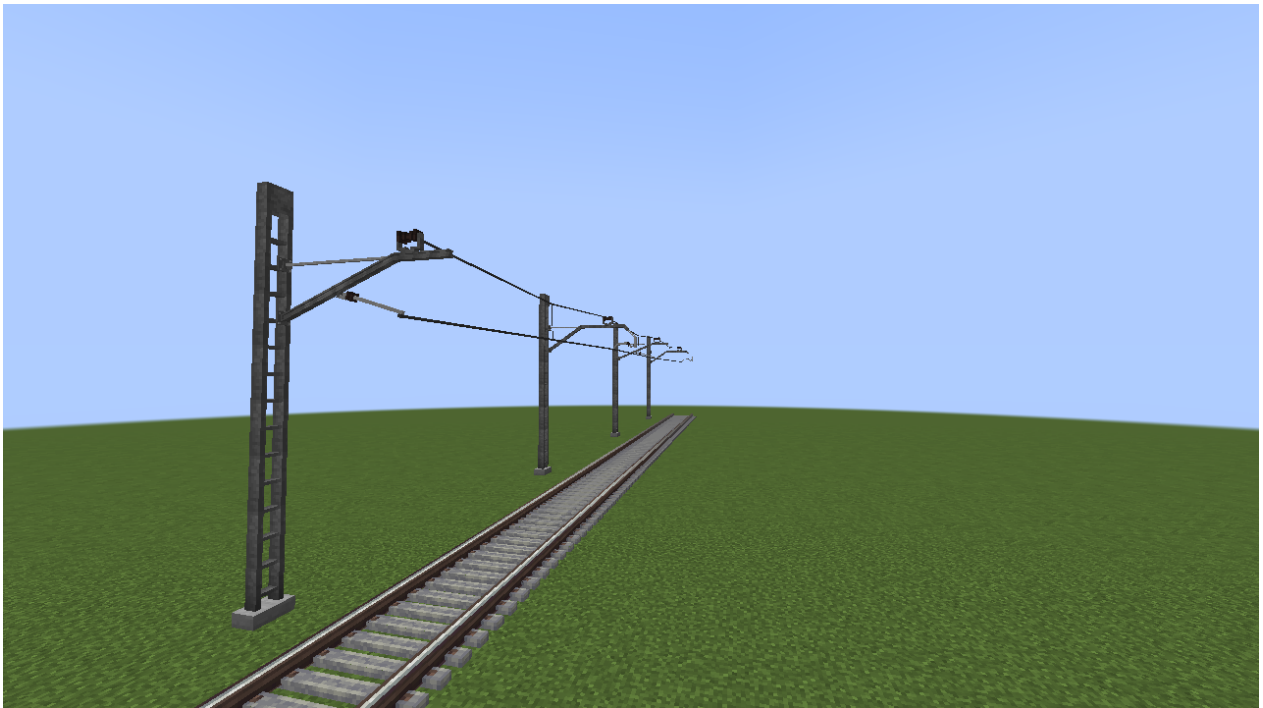


Figure 1: A stretch of line built with the add-on: Iberian lattice masts, Iberian cantilevers in zigzag and the overhead wire of *Create: Pantographs & Wires*.

Developed by a railway enthusiast.

Contents

1	Overview	2
2	Real-world reference	3
3	Iberian Cantilever	4
3.1	Zigzag position	4
3.2	Insulator type and support wire	5
3.3	Settings window	5
3.4	Placement	6
4	Iberian masts	7
5	Rigid catenary	8
5.1	Transition from flexible to rigid catenary	8
6	Iberian Tunnel Support	8
7	Iberian Pantograph	10
8	Items and crafting	10
9	Technical summary	11

1 Overview

Iberian Cantilevers is an add-on for *Create: Pantographs & Wires* (PNW) that adds overhead line equipment as it is built on the railways of the Iberian Peninsula, mostly following Spanish practice: the cantilevers (*ménsulas*) of the conventional network, the masts that carry them, the rigid overhead conductor rail used in tunnels and an Iberian-style pantograph.

The add-on does not replace any part of PNW. It plugs into PNW's wire system: its blocks are PNW wire connectors, the catenary wire of PNW is strung onto them as usual, pantographs collect current from them in the same way, and the masts share PNW's tags, so PNW blocks and Iberian blocks can be mixed freely on the same line.

Contents of the add-on

- **Iberian Cantilever** – a configurable cantilever modelled on the Spanish one, with three zigzag positions, three insulator types and an optional support wire (Section 3).
- **Iberian masts** – flat lattice (straight and diagonal) and square lattice masts with four weathering stages and waxing (Section 4).
- **Rigid catenary** – an overhead conductor rail strung like a wire, which follows curves and can be joined to the flexible catenary (Section 5).
- **Iberian Tunnel Support** – ceiling and wall supports for the rigid catenary in several sizes (Section 6).
- **Iberian Pantograph** – PNW's pantograph with the red frame typical of Iberian trains (Section 7).

2 Real-world reference

On the Spanish conventional network the contact wire is usually held by *cantilevers*: a steel tube fixed to a lattice or H-beam mast, which rises in a short diagonal and then runs horizontally over the track. The messenger wire rests on an insulator on top of the tube, and a registration arm hanging from the cantilever pulls the contact wire alternately to each side of the track (*zigzag*), so that the pantograph strip wears evenly. A support wire (stay) often runs from the mast to the elbow of the tube.



(a) Cantilevers on lattice masts, Sant Adrià de Besòs.

Adif – Ineco, CC BY-SA 3.0, via Wikimedia Commons.



(b) Cantilevers on flat lattice masts, L'Ampolla.

9pm, CC BY 4.0, via Wikimedia Commons.

Figure 2: Cantilevers of the Spanish conventional network.



Figure 3: Double-track line with cantilevers on both sides.

Javier Beristain Labaca, public domain, via Wikimedia Commons.

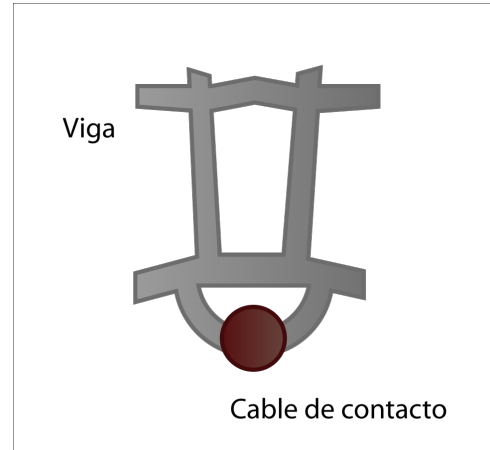
In tunnels and underground stations the flexible catenary is often replaced by a *rigid catenary*: an aluminium or steel profile that clamps the contact wire and hangs from short supports fixed to the ceiling or to the tunnel wall. It needs very little height and almost no maintenance of tension.

Many Spanish electric trains carry pantographs with a frame painted red, which is where the colour of the Iberian Pantograph comes from.



(a) Rigid catenary and its ceiling supports, Nuevos Ministerios (Madrid Metro, Line 10).

Draceane, CC BY-SA 4.0, via Wikimedia Commons.



(b) Cross-section of a rigid catenary profile (*viga*) with the contact wire (*cable de contacto*) clamped below.

Public domain, via Wikimedia Commons.

Figure 4: Rigid overhead conductor rail.



(a) Pantograph on a Renfe Class 446 train, Córdoba.

Philip Mallis, CC BY-SA 4.0, via Wikimedia Commons.



(b) Pantograph on a Renfe Class 449 train.

Falk2, CC BY-SA 4.0, via Wikimedia Commons.

Figure 5: Red pantograph frames on Spanish trains.

3 Iberian Cantilever

The Iberian Cantilever is placed on the side of a mast, exactly like PNW's cantilever, and accepts the PNW catenary wire: the messenger wire is attached on top of the insulator and the contact wire at the tip of the registration arm. All its parts are built from separate models (tube, perforated drop pieces, registration arm, insulators, support wire) that are assembled and sized automatically from its settings, so the cantilever always reaches exactly where the wires are.

3.1 Zigzag position

- **Inner** – the contact wire is held towards the mast. The registration arm hangs from the diagonal or, for wide cantilevers, from an L-shaped strut below the horizontal tube.
- **Middle** – the tube continues past the messenger insulator, drops at 45° into a perforated piece and a longer registration arm descends to the wire.

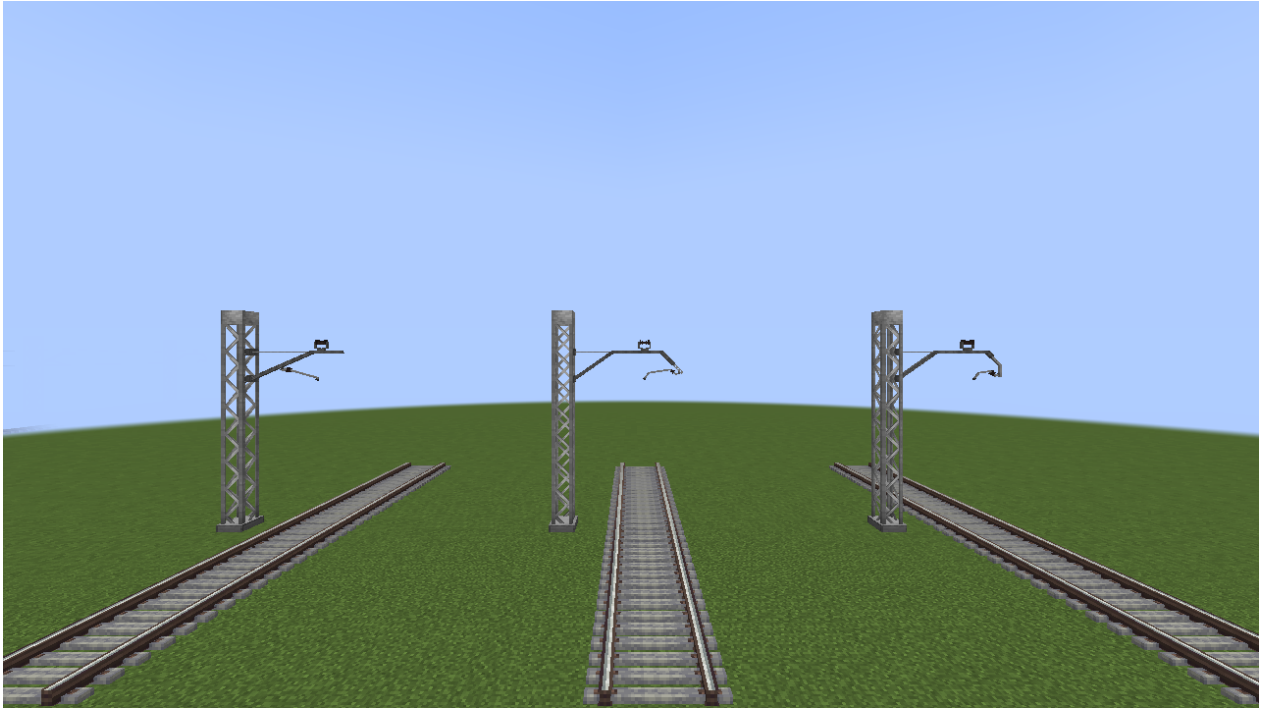
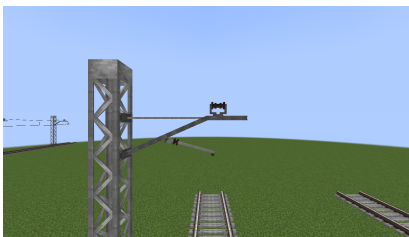
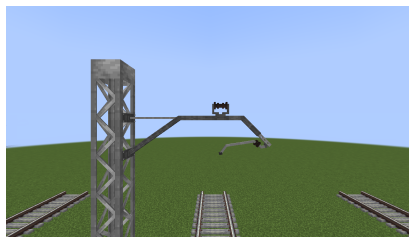


Figure 6: The three zigzag positions of the contact wire: inner, middle and outer.

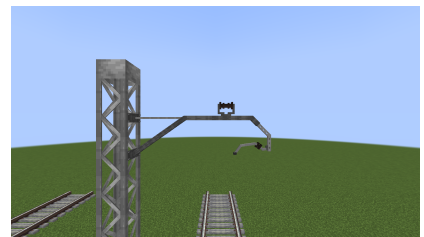
- **Outer** – the tube drops into a diagonal and a vertical perforated piece, and the registration arm points back towards the mast.



(a) Inner.



(b) Middle.



(c) Outer.

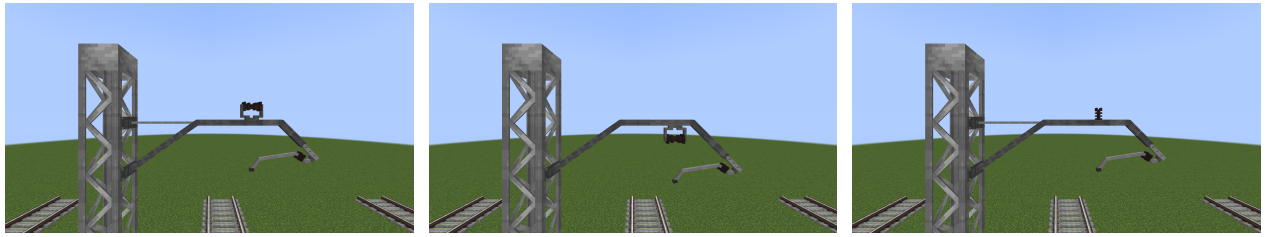
Figure 7: Side view of each zigzag position. The middle and outer positions use an arched registration arm with its own insulator.

3.2 Insulator type and support wire

The insulator that carries the messenger wire can be chosen among three types, and the support wire (stay) between the mast and the elbow of the tube can be added or removed.

3.3 Settings window

Right-clicking with the cantilever item in hand (or with an empty hand on a placed cantilever) opens a settings window in the same style as PNW's, with a live 3D preview.



(a) Horizontal support: the messenger rests on top of a horizontal insulator. (b) Horizontal hanging support: the insulator hangs below the tube (shown without support wire). (c) Vertical support: a vertical insulator on top of the tube.

Figure 8: Insulator types.



Figure 9: Settings window of the Iberian Cantilever.

Setting	Effect
Width	Reach of the cantilever from the mast to the track axis, from 1.5 to 6.5 blocks.
Insulator type	Horizontal support, horizontal hanging support or vertical support.
Contact wire	Zigzag position: inner, middle or outer.
Support wire	With or without the stay between mast and tube.
Supporter height	Drop of the diagonal of the tube (advanced).
Catenary height	Height of the contact wire below the block (advanced).
Y offset	Lowers the whole cantilever (advanced).
Set default config	Stores the current settings as the default for new, unconfigured items.

Without advanced options the heights follow fixed proportions, as PNW does with its own cantilever. Changing the settings of a cantilever that already has wires attached moves the wires with it.

3.4 Placement

- Attaches to PNW masts, Iberian masts, concrete pillars and any other block that PNW's cantilever accepts, in the same 16 directions as PNW blocks (including 45° on diagonal masts).
- The mounting plate adapts to the thickness of the mast and widens on thick masts or full blocks without deforming its bolts.
- The item stacks up to 64.

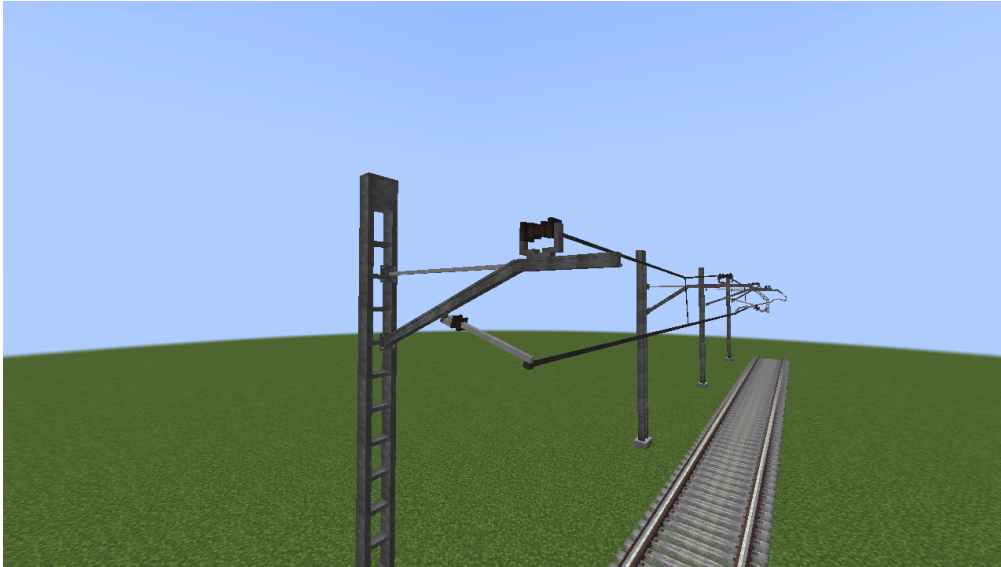


Figure 10: Cantilevers strung with PNW catenary wire: messenger, contact wire and droppers.

4 Iberian masts

Three mast families with an Iberian finish: *Iberian Flat Lattice Mast*, *Iberian Flat Diagonal Lattice Mast* and *Iberian Lattice Mast* (square section). They reuse PNW's mast behaviour:

- four weathering stages (new, exposed, weathered, oxidized) that progress over time;
- waxing with honeycomb and wax removal with an axe, as with PNW masts;
- the same connection tags as PNW masts, so both PNW's cantilevers and Iberian Cantilevers attach to them, and the flat mast can be switched between its straight and diagonal versions by crafting.

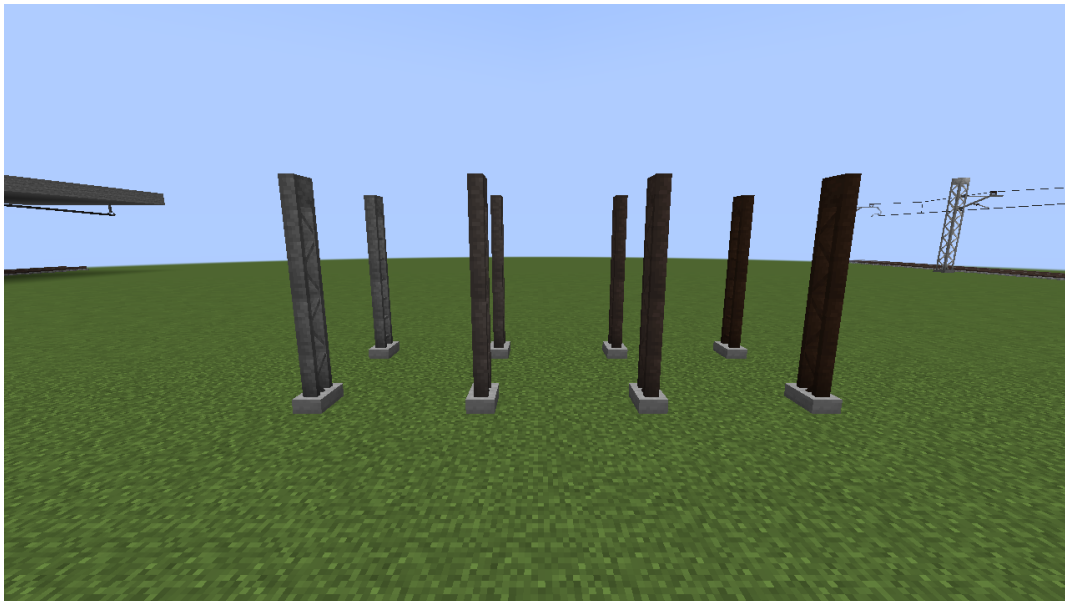
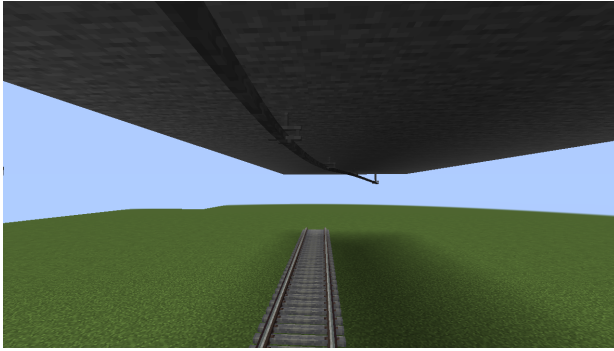


Figure 11: Iberian flat lattice masts, straight and diagonal, in their four weathering stages.

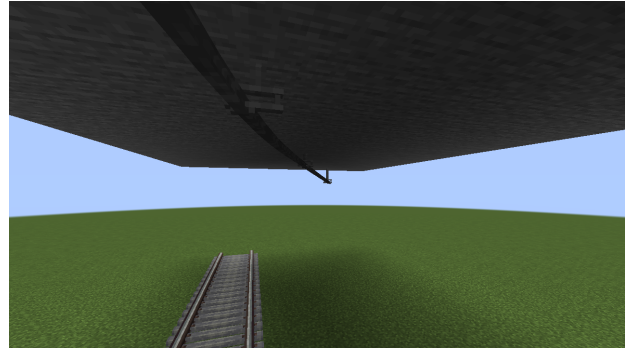
5 Rigid catenary

The *Rigid Catenary Profile Bundle* is used like PNW's wire coil: click a support, then the next one, and a rigid profile is laid between them. For PNW it is one more wire of the catenary network, so pantographs collect current from it.

- A bundle holds 200 metres of profile and is consumed by length; a bar shows what is left.
- Where two spans meet at a support with a gentle angle, the profile bends smoothly through the support instead of making a sharp corner.
- Breaking a span returns its metres to the bundles in the player's inventory; whatever does not fit is dropped as a single bundle.



(a) Rigid catenary through a tunnel.

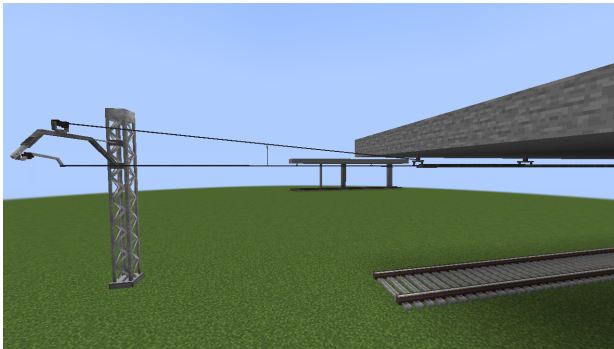


(b) The profile following a curve.

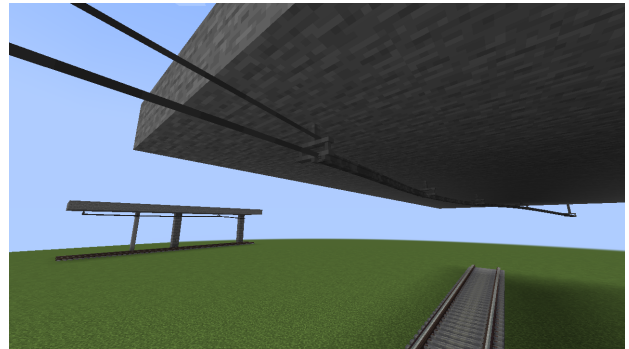
Figure 12: Rigid catenary.

5.1 Transition from flexible to rigid catenary

PNW's catenary wire can be strung from a cantilever to a tunnel support. The contact wire enters the clamp of the rigid profile and the messenger wire descends to an anchoring point just above it; at the transition support the end of the profile turns towards the incoming contact wire so that the pantograph passes smoothly. With default settings, a tunnel support placed under a cantilever keeps the contact wire level across the transition.



(a) Transition span.



(b) Contact wire and messenger at the first tunnel support.

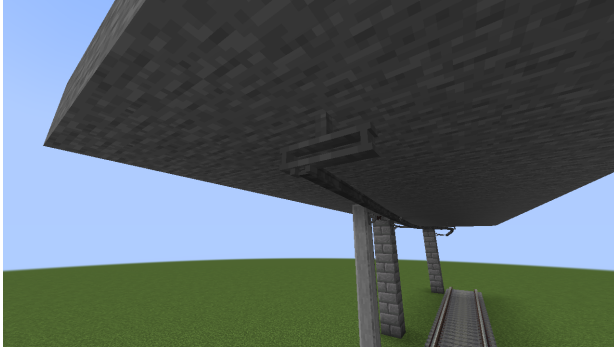
Figure 13: Transition between flexible and rigid catenary.

6 Iberian Tunnel Support

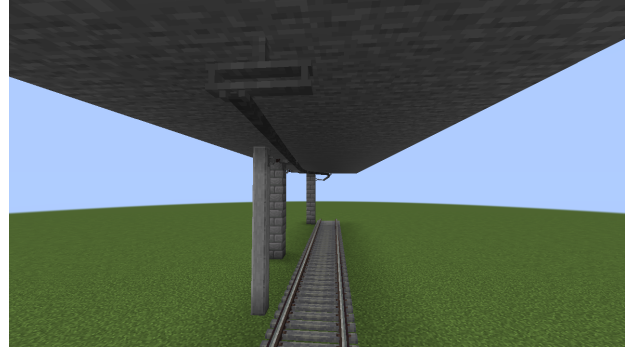
A single block covers every support of the rigid catenary. It is configured from one settings window and comes in two versions:

- **Ceiling** – hangs from the block above on a rod. Two sizes (normal and large, with cross-members spanning the whole block) and three clamp positions (left, centre, right) to build the zigzag. Its height can be increased in quarter-block steps; the rod stretches accordingly.
- **Wall** – a horizontal insulated arm fixed to a wall, to the side of a PNW mast or to a concrete pillar, placed like a cantilever in 16 directions. Two arm lengths (short and long). Its height can be moved up or down within the block in half-pixel steps, starting from the centre.

The version is chosen automatically when placing: clicking the side of a wall or pillar places the wall version, clicking the underside of a ceiling places the ceiling version.

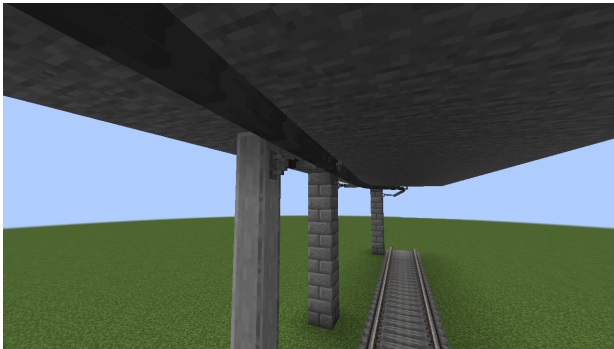


(a) Large ceiling support.

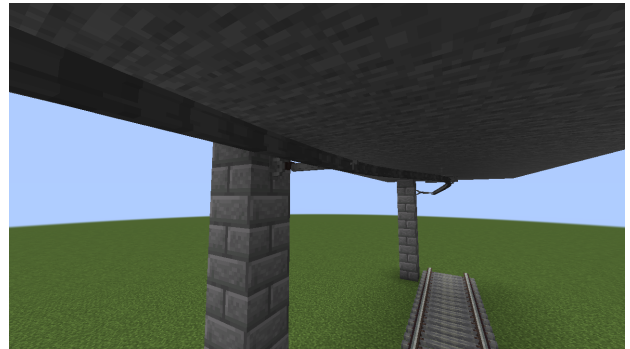


(b) Ceiling and wall supports on the same line.

Figure 14: Tunnel support variants.



(a) Short wall arm on a PNW concrete pillar.



(b) Long wall arm on a masonry wall.

Figure 15: Wall version.



(a) Ceiling version, with the ceiling block shown for reference.



(b) Wall version, with the wall block shown for reference.

Figure 16: Settings window of the tunnel support: version, size, clamp position and height, plus a button to store the current settings as the default.

7 Iberian Pantograph

The Iberian Pantograph is PNW's pantograph with its frame painted red, as on many Spanish trains. It keeps the model, animation and behaviour of the original: it can be raised and lowered, it collects current from both the flexible and the rigid catenary, and it works on Create trains and contraptions. Insulators and the contact strip keep their original colours.



Figure 17: PNW pantograph (left) and Iberian Pantograph (right).

8 Items and crafting

	Item	Recipe
	Iberian Steel	Iron ingot and black dye (shapeless). Base material of the add-on.
	Iberian Cantilever	Create sequenced assembly from a PNW iron rod bundle: two loops of deploying Iberian Steel, deploying a brown insulator and pressing.
	Iberian Tunnel Support	Two Iberian Steel above a brown insulator; makes two.
	Rigid Catenary Profile Bundle	Six Iberian Steel.
	Iberian masts	PNW iron strips with Iberian Steel (and iron rods for the square mast); six per craft.
	Iberian Pantograph	PNW pantograph and red dye (shapeless).

All recipes are visible in recipe viewers such as JEI, including the Create sequenced assembly.

9 Technical summary

Minecraft	1.20.1, Forge 47
Required mods	Create: Pantographs & Wires (1.20.1-beta-0.2.3-C6), Create 6.0.8, DragonLib, GeckoLib
License	GPL-3.0, the same as Pantographs & Wires
Languages	English, Spanish, Catalan
Mod id	iberiancantilever

Integration with Pantographs & Wires The add-on relies only on PNW's public building blocks: wire connectors, wire types registered in PNW's wire registry and catenary graph, PNW's mast classes and tags, its pantograph and its GUI widgets. Its own content (models, textures, translations and code) is self-contained, so it can be distributed as a separate add-on that requires PNW, or merged into PNW as a set of additional blocks.

Credits *Create: Pantographs & Wires* by MrJulsen, on which this add-on is built, and *Create* by the Create team. Photographs from Wikimedia Commons, credited under each image and listed below.

- *Catenaria Sant Adrià (Adif)* – Adif – Ineco, CC BY-SA 3.0. [https://commons.wikimedia.org/wiki/File:Catenaria_Sant_Adri%C3%A0_\(Adif\).jpg](https://commons.wikimedia.org/wiki/File:Catenaria_Sant_Adri%C3%A0_(Adif).jpg)
- *Catenària L'Ampolla 2023* – 9pm, CC BY 4.0. https://commons.wikimedia.org/wiki/File:Caten%C3%A0ria_L%27Ampolla_2023.jpg
- *Catenaria en España CAA* – Javier Beristain Labaca, public domain. https://commons.wikimedia.org/wiki/File:Catenaria_en_Espa%C3%B1a_CAA.JPG
- *Nuevos Ministerios L10 catenaria rígida* – Draceane, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:Nuevos_Ministerios_L10_catenaria_rigida.jpg
- *Sección catenaria rígida* – public domain. https://commons.wikimedia.org/wiki/File:Secci%C3%B3n_catenaria_r%C3%ADgida.svg
- *Pantograph on top of Renfe Class 446 train at Córdoba* – Philip Mallis, CC BY-SA 4.0. [https://commons.wikimedia.org/wiki/File:Pantograph_on_top_of_Renfe_Class_446_train_at_C%C3%B3rdoba_Julio_Anguita_Railway_Station_in_C%C3%B3rdoba,_Spain_\(55066435535\).jpg](https://commons.wikimedia.org/wiki/File:Pantograph_on_top_of_Renfe_Class_446_train_at_C%C3%B3rdoba_Julio_Anguita_Railway_Station_in_C%C3%B3rdoba,_Spain_(55066435535).jpg)
- *S01 882 Stromabnehmer BR 449* – Falk2, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:S01_882_Stromabnehmer_BR_449.jpg